# Chapter 3: Vectors and the Geometry of Space

Calculus II

February 25, 2025

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# Section 1

# Two-dimensional vectors

# Two-dimensional vectors

## Definition 3.1.1

An ordered pair of numbers is called a two-dimensional vector. We denote a vector by a capital letter:

$$\mathbf{U} = (u_1, u_2)\,.$$

- The numbers $u_1$ and $u_2$ are called the components of the vector $\mathbf{U}$.
- The set of all two-dimensional vectors, denoted $\mathbb{R}^2$, is called two-dimensional space.

# Algebraic operations for two-dimensional vectors

(a) For a vector $\mathbf{U} = (u_1, u_2)$ and a number $c \in \mathbb{R}$, define

$$c\mathbf{U} := (cu_1, cu_2)\,.$$

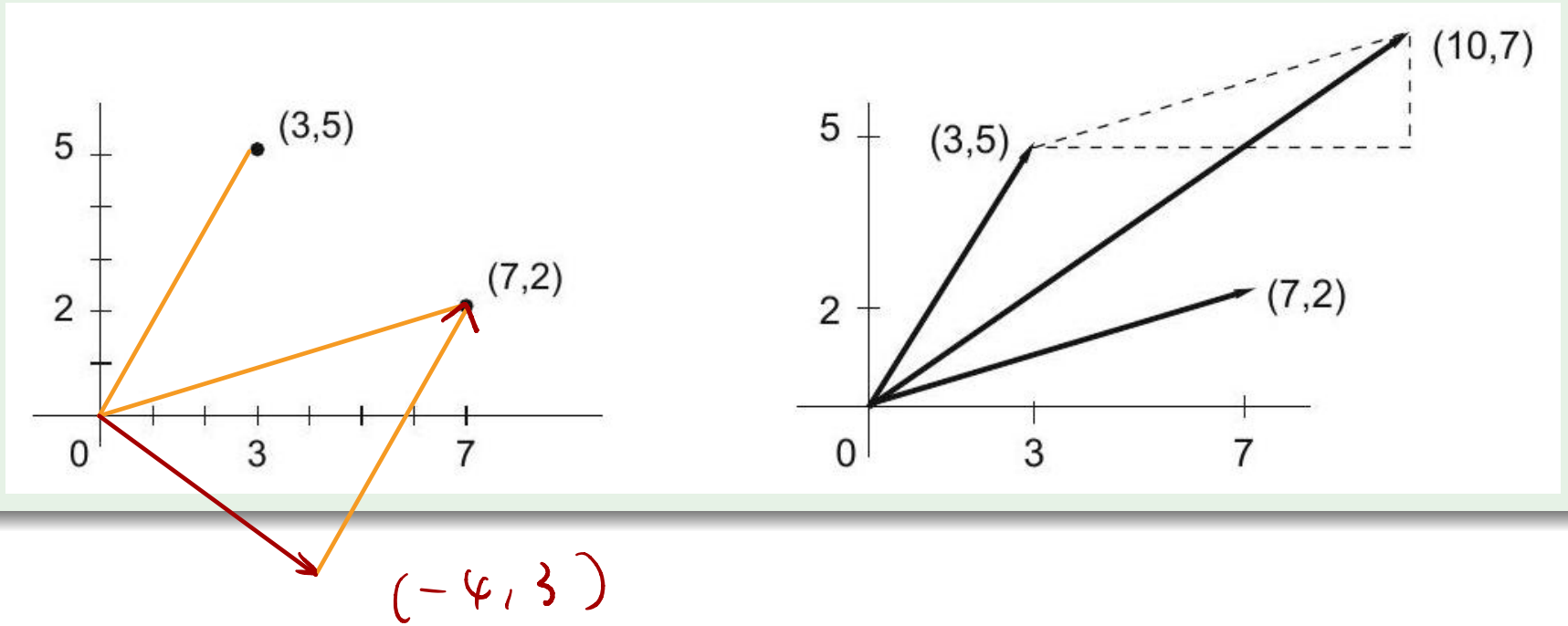
(b) For vectors $\mathbf{U} = (u_1, u_2)$ and $\mathbf{V} = (v_1, v_2)$, define

$$\mathbf{U} + \mathbf{V} = (u_1 + v_1, u_2 + v_2)\,.$$

- We denote $(0, 0)$ as $\mathbf{0}$ and call it the zero vector.
- Note that $\mathbf{U} + \mathbf{0} = \mathbf{U}$ for every vector $\mathbf{U}$.
- The symbol $-\mathbf{U}$ denotes the vector $(-u_1, -u_2)$.
- The vector $\mathbf{V} - \mathbf{U}$ defined as $\mathbf{V} + (-\mathbf{U})$ is called the difference of $\mathbf{V}$ and $\mathbf{U}$.

## Example 3.1.2

Two vectors (3,5) and (7,2) and their sum



# Usual algebraic properties

Multiplication by a number (or scalar) and addition of vectors have the usual algebraic properties:

$$\mathbf{U} + \mathbf{V} = \mathbf{V} + \mathbf{U} \qquad \text{commutative}$$

$$(\mathbf{U} + \mathbf{V}) + \mathbf{W} = \mathbf{U} + (\mathbf{V} + \mathbf{W}) \qquad \text{associative}$$

$$c(\mathbf{U} + \mathbf{V}) = c\mathbf{U} + c\mathbf{V} \qquad \text{distributive}$$

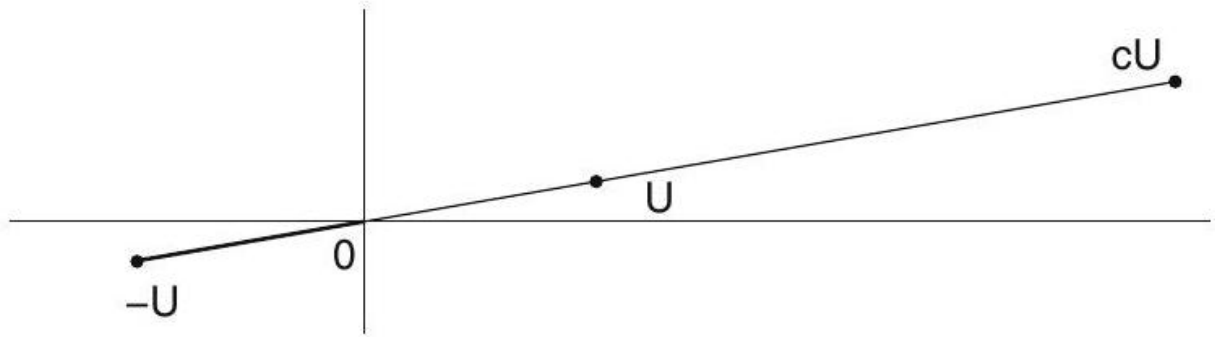
$$(a + b)\mathbf{U} = a\mathbf{U} + b\mathbf{U} \qquad \text{distributive}$$

$$\mathbf{U} + (-\mathbf{U}) = \mathbf{0} \qquad \text{additive inverse}$$

# Geometric interpretation: multiplication by a number

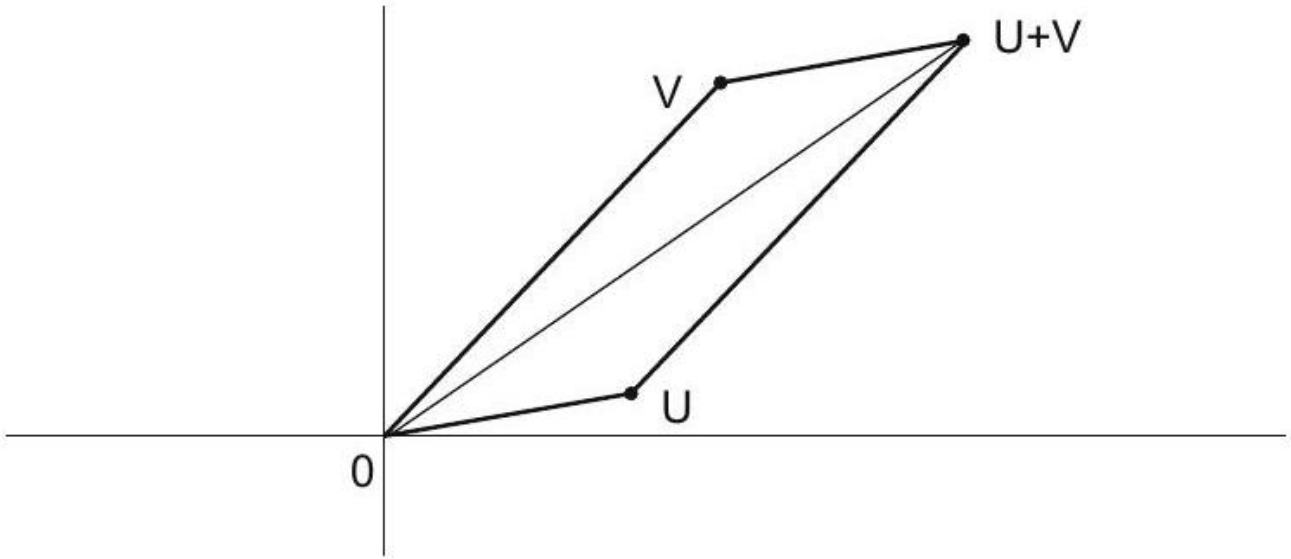
(a) For a nonzero vector $\mathbf{U}$ and a number $c$, the point $c\mathbf{U}$ lies on the line through the origin and the point $\mathbf{U}$.

- Its distance from the origin is $|c|$ times the distance of $\mathbf{U}$ from the origin.
- The origin divides this line into two rays; when $c$ is positive, $\mathbf{U}$ and $c\mathbf{U}$ lie on the same ray; when $c$ is negative, $\mathbf{U}$ and $c\mathbf{U}$ lie on opposite rays.



# Geometric interpretation: addition of two vectors

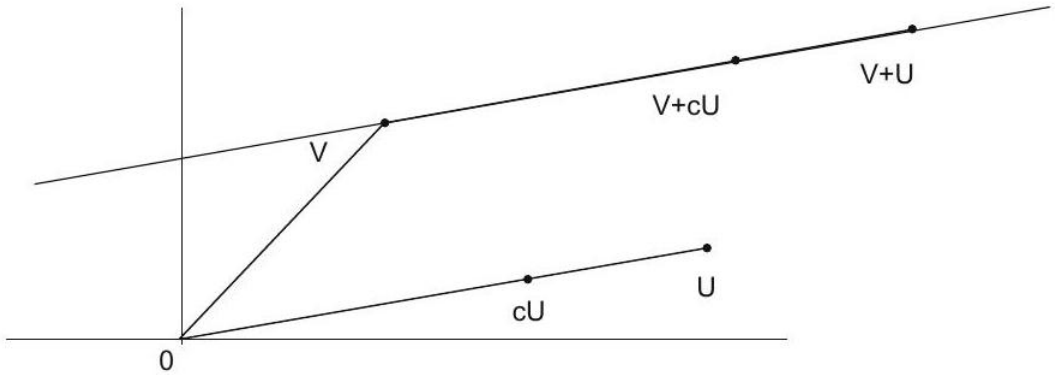
(b) If the points $\mathbf{0}, \mathbf{U}$, and $\mathbf{V}$ do not lie on a line, the four points $\mathbf{0}, \mathbf{U}, \mathbf{V}$, and $\mathbf{U} + \mathbf{V}$ form the vertices of parallelogram.



# Geometric interpretation

(c) For $c$ between 0 and 1 the points $\mathbf{V} + c\mathbf{U}$ lie on the line segment from $\mathbf{V}$ to $\mathbf{V} + \mathbf{U}$.

- That side of the parallelogram is parallel to the segment from $\mathbf{0}$ to $\mathbf{U}$ and has the same length.
- The directed line segment from $\mathbf{V}$ to $\mathbf{V} + \mathbf{U}$ is another way to visualize the vector $\mathbf{U}$.




**Remark:** We can visualize vector addition and multiplication by a number in *two dimensions*. But we will see that in dimensions higher than three it is the *algebraic properties* of vectors that are most useful.

# Section 2

## The norm and dot product of two-dimensional vectors

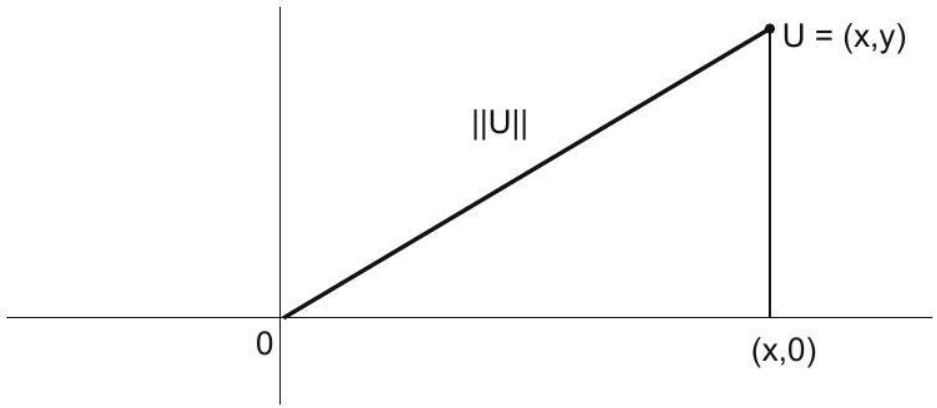
# The norm of vector

## Definition 3.2.1

The norm of $\mathbf{U} = (x, y)$, denoted as $\|\mathbf{U}\|$, is defined as

$$\|\mathbf{U}\| = \sqrt{x^2 + y^2}.$$

A *unit vector* is a vector of norm 1.




- Applying the Pythagorean theorem to the triangle whose vertices are $(0, 0)$, $(x, 0)$, and $(x, y)$, we see that the norm of $(x, y)$ is the distance between $(x, y)$ and the origin.
- The norm of $\mathbf{U}$ is also sometimes called the length of $\mathbf{U}$.

## Example 3.2.2

The norm of $\mathbf{U} = (1, 2)$ is

$$\|\mathbf{U}\| = \sqrt{1^2 + 2^2} = \sqrt{5}$$

The norm of $\mathbf{V} = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ is

$$\|\mathbf{V}\| = \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2} = 1$$

so $\mathbf{V}$ is a unit vector.

# The dot product of vector

A concept related to norm is the dot product of two vectors $\mathbf{U}$ and $\mathbf{V}$.

## Definition 3.2.3

The dot product of $\mathbf{U} = (u_1, u_2)$ and $\mathbf{V} = (v_1, v_2)$ is

$$\mathbf{U} \cdot \mathbf{V} = u_1 v_1 + u_2 v_2$$

# Properties of the dot product

The dot product has some of the usual properties of a product:
(a) Distributive: for vectors $\mathbf{U}, \mathbf{V}$, and $\mathbf{W}$

$$\mathbf{U} \cdot (\mathbf{V} + \mathbf{W}) = \mathbf{U} \cdot \mathbf{V} + \mathbf{U} \cdot \mathbf{W}$$

(b) Commutative: for vectors $\mathbf{U}$ and $\mathbf{V}$

$$\mathbf{U} \cdot \mathbf{V} = \mathbf{V} \cdot \mathbf{U}$$

# Properties of the dot product

Note that for a vector $\mathbf{U}$,

$$\mathbf{U} \cdot \mathbf{U} = \|\mathbf{U}\|^2.$$

By distributive and commutative laws, for vectors $\mathbf{U}$ and $\mathbf{V}$:

$$(\mathbf{U}-\mathbf{V})\cdot(\mathbf{U}-\mathbf{V}) = \mathbf{U}\cdot(\mathbf{U}-\mathbf{V}) - \mathbf{V}\cdot(\mathbf{U}-\mathbf{V}) = \mathbf{U}\cdot\mathbf{U} - \mathbf{U}\cdot\mathbf{V} - \mathbf{V}\cdot\mathbf{U} + \mathbf{V}\cdot\mathbf{V},$$

which can be rewritten as

$$\|\mathbf{U}-\mathbf{V}\|^2 = \|\mathbf{U}\|^2 + \|\mathbf{V}\|^2 - 2\mathbf{U}\cdot\mathbf{V} \tag{1}$$

Since $\|\mathbf{U}-\mathbf{V}\|^2$ is nonnegative, it follows that

$$\mathbf{U}\cdot\mathbf{V} \leq \frac{1}{2}\|\mathbf{U}\|^2 + \frac{1}{2}\|\mathbf{V}\|^2. \tag{2}$$

We show next that an even sharper inequality holds.

# Cauchy-Schwarz inequality

## Theorem 3.2.4 (Cauchy-Schwarz inequality)

*For all vectors* $\mathbf{U}$ *and* $\mathbf{V}$ *in* $\mathbb{R}^2$ *the following inequality holds:*

$$|\mathbf{U} \cdot \mathbf{V}| \leq \|\mathbf{U}\|\|\mathbf{V}\|. \tag{3}$$

**Proof:** If either $\mathbf{U}$ or $\mathbf{V}$ is the zero vector, inequality (3) holds, because both sides are zero.

For all nonzero vectors $\mathbf{U}$ and $\mathbf{V}$,

$$\frac{1}{\|\mathbf{U}\|}\mathbf{U}, \quad \frac{1}{\|\mathbf{V}\|}\mathbf{V}$$

are unit vectors and therefore by inequality (2)

$$\frac{\mathbf{U} \cdot \mathbf{V}}{\|\mathbf{U}\|\|\mathbf{V}\|} \leq 1 \quad \text{and} \quad \mathbf{U} \cdot \mathbf{V} \leq \|\mathbf{U}\|\|\mathbf{V}\|.$$

Applying the last inequality to $-\mathbf{U}$ and $\mathbf{V}$ we get

$$-\mathbf{U} \cdot \mathbf{V} \leq \|-\mathbf{U}\|\|\mathbf{V}\| = \|\mathbf{U}\|\|\mathbf{V}\|.$$

Combining the last two inequalities we obtain (3).

# Properties of the dot product

For all vectors $\mathbf{U}$ and $\mathbf{V}$,

$$0 \leq (\|\mathbf{U}\| - \|\mathbf{V}\|)^2 = \|\mathbf{U}\|^2 + \|\mathbf{V}\|^2 - 2\|\mathbf{U}\|\|\mathbf{V}\|.$$

Therefore

$$\|\mathbf{U}\|\|\mathbf{V}\| \leq \frac{1}{2}\|\mathbf{U}\|^2 + \frac{1}{2}\|\mathbf{V}\|^2$$

By Cauchy-Schwarz inequality we see that

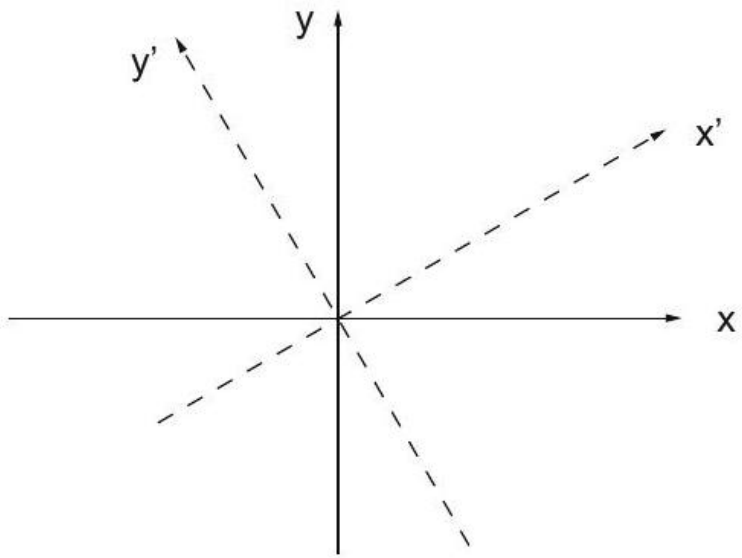
$$|\mathbf{U} \cdot \mathbf{V}| \leq \|\mathbf{U}\|\|\mathbf{V}\| \leq \frac{1}{2}\|\mathbf{U}\|^2 + \frac{1}{2}\|\mathbf{V}\|^2$$

# Dot product v.s. coordinate axes

Suppose we replace the coordinate axes by another pair of perpendicular lines through the origin. See the below figure. Let $x'$ and $y'$ be the coordinates in the new system of a vector $\mathbf{U}$ whose coordinates in the original system were $x$ and $y$. Then

$$x'^2 + y'^2 = x^2 + y^2$$

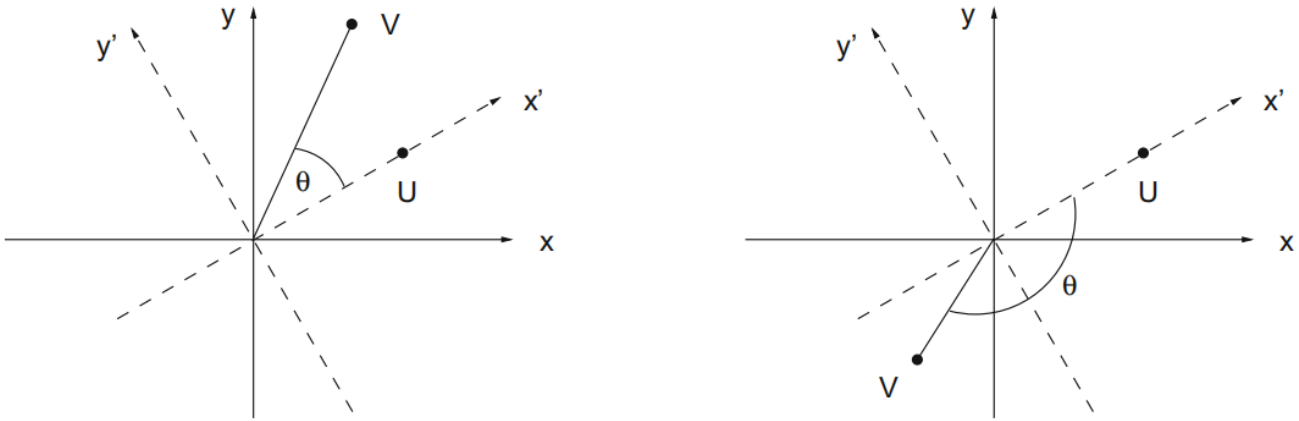
because both sides express the square of the distance between the point $\mathbf{U}$ and the origin.




The dot product of two vectors calculated in the new coordinates is equal to their dot product calculated in the old coordinates. To see that we note that formula (1) holds in both coordinate systems. The term on the left and the first two terms on the right are norms and therefore have the same value in both coordinate systems; but then the remaining term on the right, two times the dot product of $\mathbf{U}$ and $\mathbf{V}$, must also have the same value in both coordinate systems.

# Cont'd

We now use a coordinate system where the new positive $x'$ axis is the ray through the origin and the point $\mathbf{U}$.




In this new coordinate system the coordinates of $\mathbf{U}$ and $\mathbf{V}$ are:

$$\mathbf{U} = (\|\mathbf{U}\|, 0), \quad \mathbf{V} = (\|\mathbf{V}\| \cos\theta, \|\mathbf{V}\| \sin\theta)$$

where $\theta \in [0, \pi]$ is the angle between $\mathbf{U}$ and $\mathbf{V}$.

So in this coordinate system the dot product of $\mathbf{U}$ and $\mathbf{V}$ is

$$\mathbf{U} \cdot \mathbf{V} = \|\mathbf{U}\| \|\mathbf{V}\| \cos\theta$$

# The properties of the dot product

Since the dot product in our two coordinate systems is the same, this proves the following theorem.

## Theorem 3.2.5

*The dot product of two nonzero vectors* $\mathbf{U}$ *and* $\mathbf{V}$ *is the product of the norms of the two vectors times the cosine of the angle between the two vectors,*

$$\mathbf{U} \cdot \mathbf{V} = \|\mathbf{U}\|\|\mathbf{V}\| \cos\theta$$

- In particular if two nonzero vectors $\mathbf{U}$ and $\mathbf{V}$ are **perpendicular**, $\theta = \frac{\pi}{2}$, their dot product is zero, and conversely.
- When the dot product of vectors $\mathbf{U}$ and $\mathbf{V}$ is zero we say that $\mathbf{U}$ and $\mathbf{V}$ are **orthogonal**.

# Section 3

## $N$-dimensional vectors

# $N$-dimensional vectors

We extend the concepts of vectors and their algebra from two dimensions to $n$ dimensions, where $n$ is any positive integer.

## Definition 3.3.1

An ordered $n$-tuple 数组.

$$\mathbf{U} = (u_1, u_2, \ldots, u_n)$$

of numbers is called an $n$-dimensional vector. The numbers $u_j$ are called the components of the vector $\mathbf{U}$, and $u_j$ is called the $j$-th component of $\mathbf{U}$. The set of all $n$-dimensional vectors, denoted $\mathbb{R}^n$, is called $n$-dimensional space.

# $N$-dimensional vectors

The vector all of whose components are zero is called the zero vector and is denoted as $\mathbf{0}$:

$$\mathbf{0} = (0, 0, \ldots, 0)$$

(a) Let $\mathbf{U} = (u_1, u_2, \ldots, u_n)$ and let $c$ be a number. Define $c\mathbf{U}$ as

$$c\mathbf{U} = (cu_1, cu_2, \ldots, cu_n)$$

(b) The sum of $\mathbf{U} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{V} = (v_1, v_2, \ldots, v_n)$ is defined as

$$\mathbf{U} + \mathbf{V} = (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n)$$

We can easily verify that $\mathbb{R}^n$ has the usual algebraic properties

$$c(\mathbf{X} + \mathbf{Y}) = c\mathbf{X} + c\mathbf{Y}, \quad \mathbf{X} + \mathbf{Y} = \mathbf{Y} + \mathbf{X}, \quad \mathbf{X} + (\mathbf{Y} + \mathbf{Z}) = (\mathbf{X} + \mathbf{Y}) + \mathbf{Z}$$

According to the third of these we write $\mathbf{X} + (\mathbf{Y} + \mathbf{Z})$ as $\mathbf{X} + \mathbf{Y} + \mathbf{Z}$.

# Section 4

## Norm and dot product in $n$ dimensions

We define now, in analogy with two-dimensional vectors, the norm of a vector in $\mathbb{R}^n$, and the dot product of two vectors in $\mathbb{R}^n$.

## Definition 3.4.1

The norm of a vector $\mathbf{U} = (u_1, u_2, \ldots, u_n)$ is defined as

$$\|\mathbf{U}\| = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}$$

The zero vector $\mathbf{0}$ has norm zero, and conversely only the zero vector has norm zero. As in $\mathbb{R}^2$ we think of the norm of $\mathbf{U}$ as the length of $\mathbf{U}$ or the distance from the origin to the point $\mathbf{U}$.

## Definition 3.4.2

We define the dot product, $\mathbf{U} \cdot \mathbf{V}$, of $\mathbf{U} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{V} = (v_1, v_2, \ldots, v_n)$ as

$$\mathbf{U} \cdot \mathbf{V} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

We ask you to verify that the dot product is distributive and commutative:

$$\mathbf{U} \cdot (\mathbf{V} + \mathbf{W}) = \mathbf{U} \cdot \mathbf{V} + \mathbf{U} \cdot \mathbf{W}, \quad \mathbf{U} \cdot \mathbf{V} = \mathbf{V} \cdot \mathbf{U}$$

The dot product of a vector with itself is its norm squared:

$$\mathbf{U} \cdot \mathbf{U} = \|\mathbf{U}\|^2$$

The relation in $\mathbb{R}^2$

$$\|\mathbf{U}-\mathbf{V}\|^2 = \|\mathbf{U}\|^2 + \|\mathbf{V}\|^2 - 2\mathbf{U}\cdot\mathbf{V} \tag{4}$$

was derived using only the distributive and commutative rules for the dot product. Therefore it holds for vectors in $\mathbb{R}^n$ as well.

Next we prove a very useful inequality that compares $\mathbf{U}\cdot\mathbf{V}$ to $\|\mathbf{U}\|\|\mathbf{V}\|$.

# The Cauchy-Schwarz inequality

## Theorem 3.4.3 (Cauchy-Schwarz inequality)

*Let* $\mathbf{U}$ *and* $\mathbf{V}$ *be vectors in* $\mathbb{R}^n$. *Then*

$$|\mathbf{U} \cdot \mathbf{V}| \leq \|\mathbf{U}\| \|\mathbf{V}\|$$

## Example 3.4.4

We have

$$(1,1,0)\cdot(0,1,1)=1, \quad (1,-1,0)\cdot(0,1,1)=-1$$

and in both cases the absolute value of the dot product is less than the product of norms, $\sqrt{2}\sqrt{2}$.

# Proof

If $\mathbf{U}$ is the zero vector then both sides of the inequality are zero, so it holds in that case. Since the square is nonnegative

$$0 \leq \|\mathbf{V} - (\mathbf{U} \cdot \mathbf{V})\mathbf{U}\|^2$$

Using (4) rewrite the right side of last inequality to get

$$\begin{aligned} 0 &\leq \|\mathbf{V}\|^2 + \|(\mathbf{U} \cdot \mathbf{V})\mathbf{U}\|^2 - 2(\mathbf{U} \cdot \mathbf{V})^2 \\ &= \|\mathbf{V}\|^2 + (\mathbf{U} \cdot \mathbf{V})^2\|\mathbf{U}\|^2 - 2(\mathbf{U} \cdot \mathbf{V})^2 \end{aligned}$$

If $\mathbf{U}$ is a unit vector, $\|\mathbf{U}\| = 1$ and we get

$$0 \leq \|\mathbf{V}\|^2 - (\mathbf{U} \cdot \mathbf{V})^2$$

so

$$(\mathbf{U} \cdot \mathbf{V})^2 \leq \|\mathbf{V}\|^2.$$

# Proof (Continue)

Taking square roots we get that for unit vectors $\mathbf{U}$,

$$|\mathbf{U} \cdot \mathbf{V}| \leq \|\mathbf{V}\|.$$

Next if $\mathbf{U}$ is a nonzero vector, $\frac{1}{\|\mathbf{U}\|}\mathbf{U}$ is a unit vector, so

$$\left| \frac{1}{\|\mathbf{U}\|}\mathbf{U} \cdot \mathbf{V} \right| \leq \|\mathbf{V}\|.$$

Multiplying by $\|\mathbf{U}\|$ we get

$$|\mathbf{U} \cdot \mathbf{V}| \leq \|\mathbf{U}\|\|\mathbf{V}\|.$$

# Triangle inequality

An important consequence of the Cauchy-Schwarz inequality is the triangle inequality for vectors in $\mathbb{R}^n$.

## Theorem 3.4.5 (Triangle inequality)

*If* $\mathbf{U}$ *and* $\mathbf{V}$ *are vectors in* $\mathbb{R}^n$ *then*

$$\|\mathbf{U}+\mathbf{V}\| \leq \|\mathbf{U}\| + \|\mathbf{V}\|$$

**Proof:** Recall that

$$\|\mathbf{U}-\mathbf{V}\|^2 = \|\mathbf{U}\|^2 + \|\mathbf{V}\|^2 - 2\mathbf{U}\cdot\mathbf{V}$$

Replace $\mathbf{V}$ by $-\mathbf{V}$ and use the Cauchy-Schwarz inequality to get

$$\begin{aligned}\|\mathbf{U}+\mathbf{V}\|^2 &= \|\mathbf{U}\|^2 + \|\mathbf{V}\|^2 + 2\mathbf{U}\cdot\mathbf{V} \\ &\leq \|\mathbf{U}\|^2 + \|\mathbf{V}\|^2 + 2\|\mathbf{U}\|\|\mathbf{V}\| = (\|\mathbf{U}\| + \|\mathbf{V}\|)^2\end{aligned}$$

Take the square root to get $\|\mathbf{U}+\mathbf{V}\| \leq \|\mathbf{U}\| + \|\mathbf{V}\|$ as asserted.

Using the Cauchy-Schwarz inequality we can define the angle between two nonzero vectors $\mathbf{U}$ and $\mathbf{V}$ in $\mathbb{R}^n$ as follows. From the Cauchy-Schwarz inequality we see that for $\mathbf{U}$ and $\mathbf{V}$ nonzero

$$-1 \leq \frac{\mathbf{U} \cdot \mathbf{V}}{\|\mathbf{U}\|\|\mathbf{V}\|} \leq 1$$

We define the angle $\theta$ between $\mathbf{U}$ and $\mathbf{V}$ to be

$$\theta = \cos^{-1}\left(\frac{\mathbf{U} \cdot \mathbf{V}}{\|\mathbf{U}\|\|\mathbf{V}\|}\right), \quad 0 \leq \theta \leq \pi$$

or $\cos\theta = \frac{\mathbf{U} \cdot \mathbf{V}}{\|\mathbf{U}\|\|\mathbf{V}\|}$.

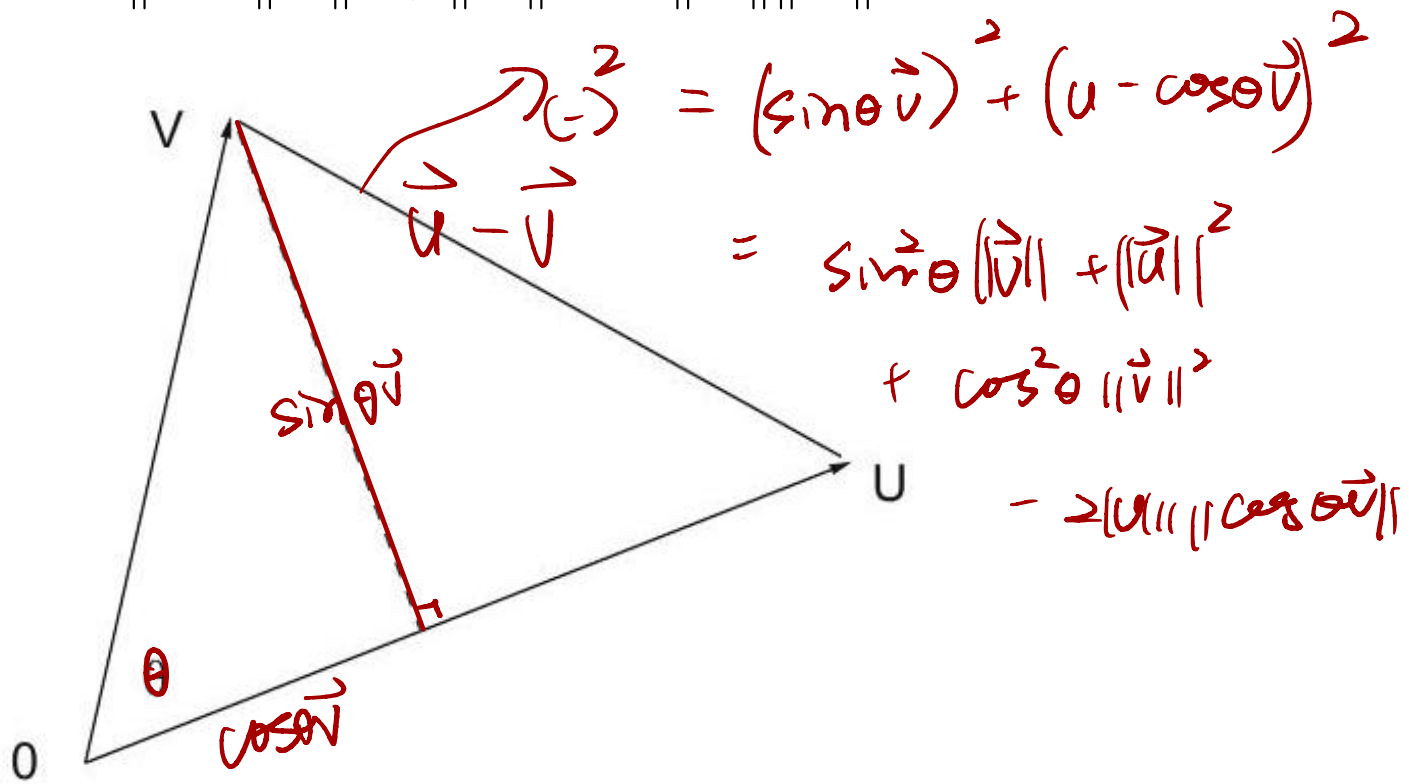
# Cont'd

Using this definition we can rewrite the formula

$$\|\mathbf{U}-\mathbf{V}\|^2 = \|\mathbf{U}\|^2 + \|\mathbf{V}\|^2 - 2\mathbf{U}\cdot\mathbf{V}$$

as

$$\|\mathbf{U}-\mathbf{V}\|^2 = \|\mathbf{U}\|^2 + \|\mathbf{V}\|^2 - 2\|\mathbf{U}\|\|\mathbf{V}\|\cos\theta$$



# Open ball

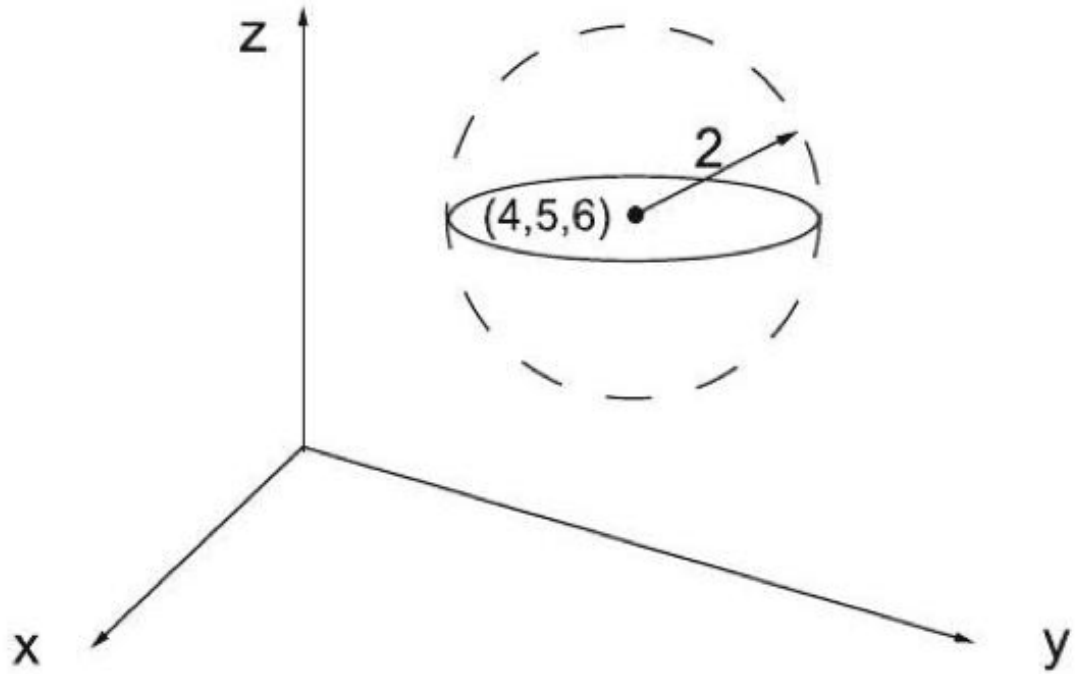
## Definition 3.4.6

An open ball of radius $r > 0$ in $\mathbb{R}^n$ centered at $\mathbf{A}$ is the set of all $\mathbf{X}$ in $\mathbb{R}^n$ with

$$\|\mathbf{X} - \mathbf{A}\| < r$$

## Example 3.4.7

The open ball of radius 2 centered at $(4, 5, 6)$ is shown in below figure. It includes all the points inside the sphere surface, but not the points on the surface, as we indicate using dotted lines.



# Interior point

## Definition 3.4.8

1. A point $\mathbf{A}$ in $D \subset \mathbb{R}^n$ is called an interior point of $D$ if there is an open ball centered at $\mathbf{A}$ that is contained in $D$.
2. The interior of $D$ is the set of interior points of $D$.

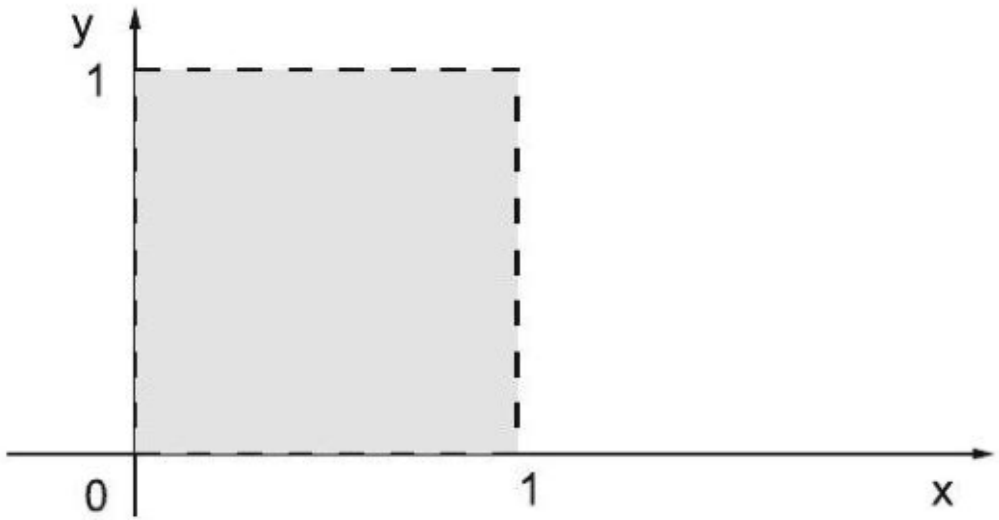
## Example 3.4.9

Take $S$ to be the square region in $\mathbb{R}^2$ consisting of all points $(x, y)$ where $0 < x < 1$ and $0 < y < 1$. Show that every point of $S$ is an interior point of $S$.

*Proof.* Let $r$ be the smallest of

$$x, \quad y, \quad 1 - x, \quad 1 - y$$

Then the open disk of radius $r$ centered at $(x, y)$ is contained in $S$, so $(x, y)$ is an interior point.



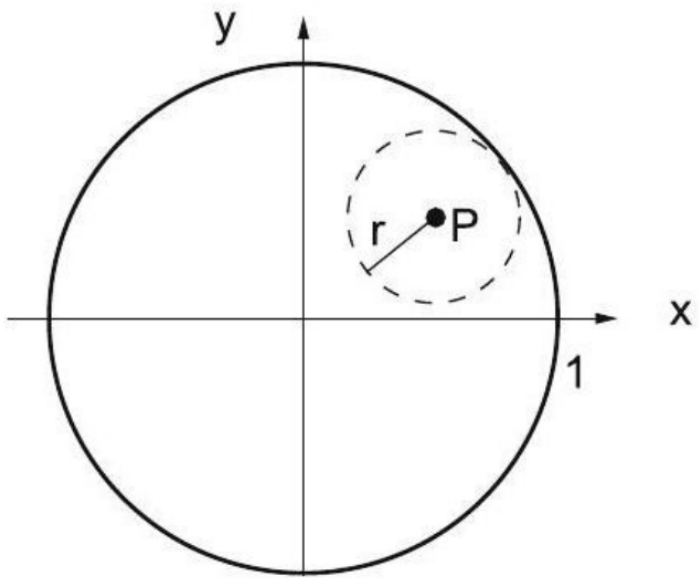
## Example 3.4.10

Take $D$ to be the unit disk $x^2 + y^2 \leq 1$ centered at the origin in the $x, y$ plane. Show that every point $\mathbf{P}$ whose distance from the origin is less than 1 is an interior point of $D$

*Proof.* Let $r = 1 - \|\mathbf{P}\|$ and let $\mathbf{Q}$ be a point in the open disk of radius $r$ centered at $\mathbf{P}$. By the triangle inequality

$$\begin{aligned} \|\mathbf{Q}\| = \|\mathbf{Q} - \mathbf{P} + \mathbf{P}\| &\leq \|\mathbf{Q} - \mathbf{P}\| + \|\mathbf{P}\| \\ &< r + \|\mathbf{P}\| = 1 \end{aligned}$$

The points that are within $r$ of $\mathbf{P}$ are all in $D$. So $\mathbf{P}$ is an interior point.



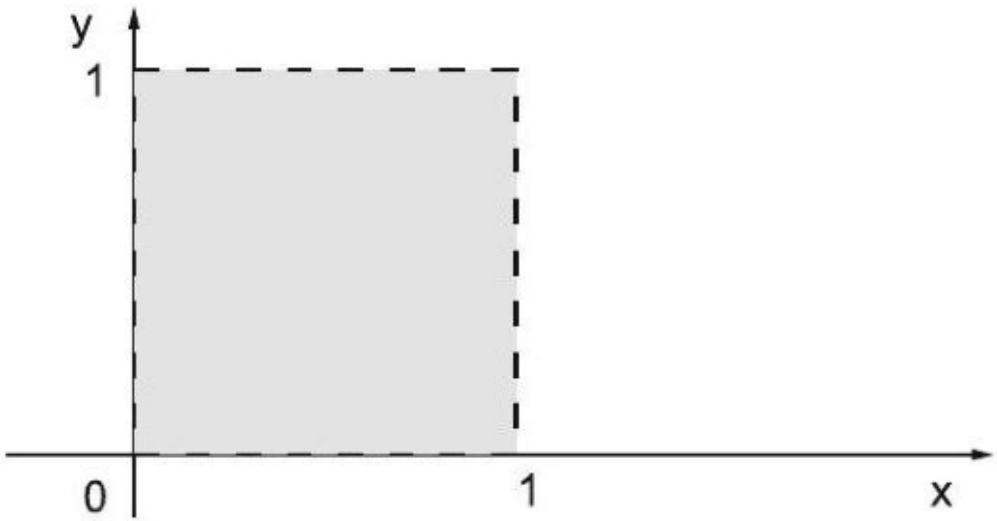
# Open sets

## Definition 3.4.11

A set $D$ in $\mathbb{R}^n$ is said to be open if every point in $D$ is an interior point.

## Example 3.4.12

Take $S$ to be the square region in $\mathbb{R}^2$ consisting of all points $(x, y)$ where $0 < x < 1$ and $0 < y < 1$. $S$ is an open set since every point of $S$ is an interior point.



# Closed sets and boundary points

## Definition 3.4.13

1. A point $\mathbf{B}$ of a set $D$ in $\mathbb{R}^n$ is a boundary point of $D$ if every ball centered at $\mathbf{B}$ contains points that are in $D$ and also points that are not in $D$.
2. The boundary of $D$ is the set of boundary points of $D$, denoted $\partial D$.

It follows from this definition that a set and its complement have the same boundary points.

## Definition 3.4.14

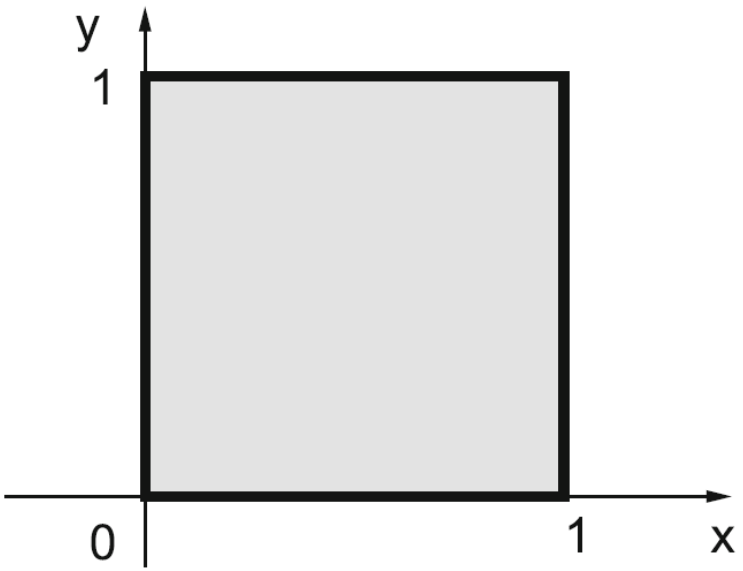
A set $D$ is called closed if it contains all its boundary points.

## Example 3.4.15

Take $S$ to be the square region

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

Every point with $x = 0$ or $x = 1$ or $y = 0$ or $y = 1$ is a boundary point. Because $S$ contains all its boundary points, $S$ is a closed set.



# Section 5

# Geometric applications

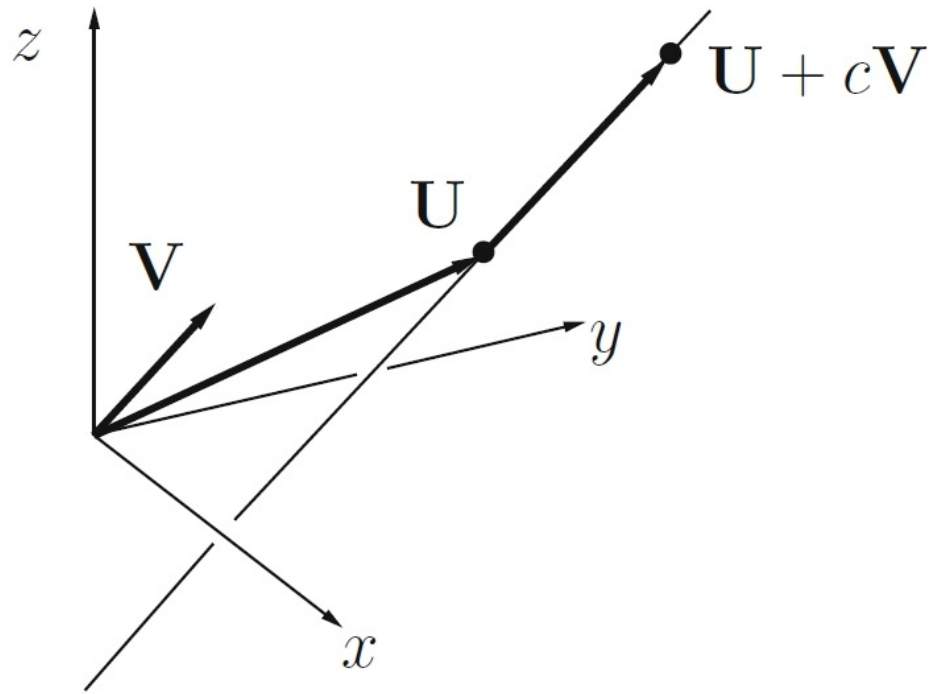
# Lines

- For two vectors $\mathbf{U}$ and $\mathbf{V}$ in $\mathbb{R}^2$, and all numbers $c$, the points

$$\mathbf{U} + c\mathbf{V}$$

  lie along a line through $\mathbf{U}$ and parallel to $\mathbf{V}$.

- If $\mathbf{U}$ and $\mathbf{V}$ are vectors in $\mathbb{R}^n$, the same expression describes a line in $\mathbb{R}^n$. See below figure for the case $n = 3$.

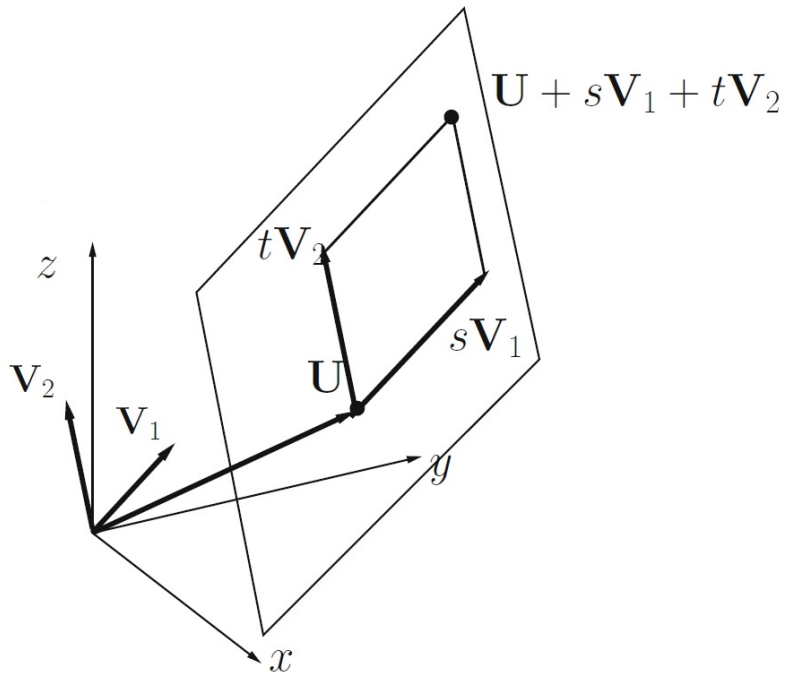


# Planes

- Let $\mathbf{U}, \mathbf{V}_1$, and $\mathbf{V}_2$ be three vectors in $\mathbb{R}^3$, where $\mathbf{V}_1$ and $\mathbf{V}_2$ are not parallel, and let $s$ and $t$ be numbers.
- For each of the points $\mathbf{U} + s\mathbf{V}_1$ on the line though $\mathbf{U}$ parallel to $\mathbf{V}_1$, there is a line $\mathbf{U} + s\mathbf{V}_1 + t\mathbf{V}_2$ through $\mathbf{U} + s\mathbf{V}_1$ parallel to $\mathbf{V}_2$.
- The set of points

$$\mathbf{U} + s\mathbf{V_1} + t\mathbf{V_2}$$

is a plane in $\mathbb{R}^3$.

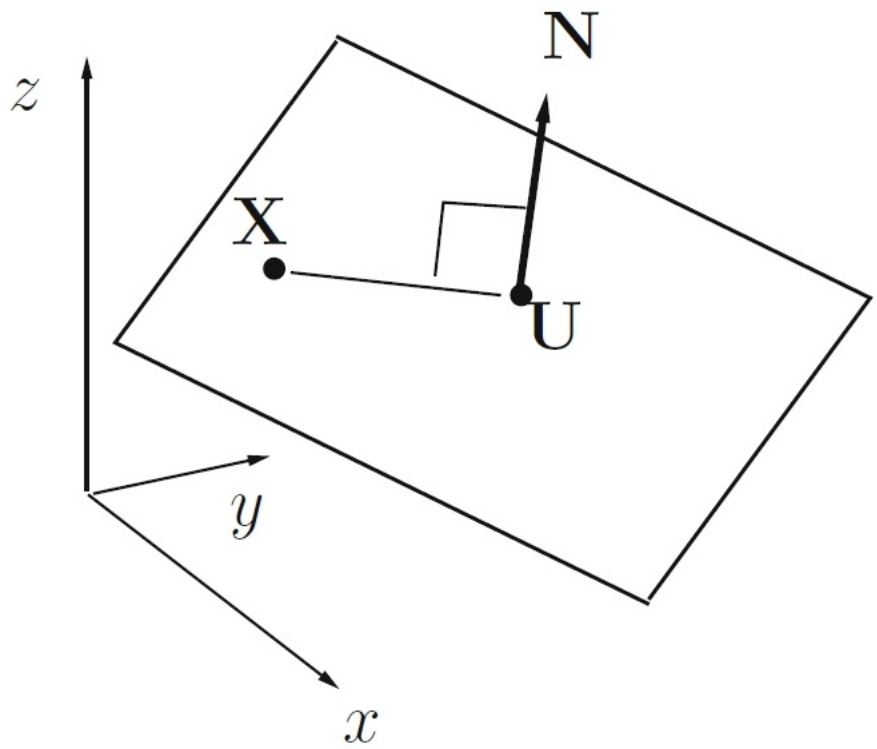


# Cont'd

In the case of $\mathbb{R}^3$ there is an alternate description for a plane, given by specifying a normal vector $\mathbf{N}$ perpendicular to the plane.

Points $\mathbf{X} = (x, y, z)$ on a plane satisfy

$$\mathbf{N} \cdot (\mathbf{X} - \mathbf{U}) = 0.$$



## Example 3.5.1

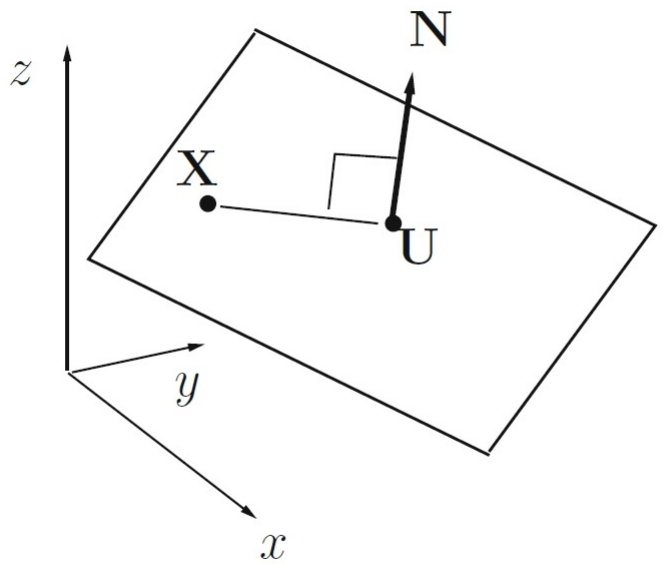
Let $\mathbf{U} = (1, 2, 3)$ and $\mathbf{N} = (2, 6, -3)$, and consider the plane containing $\mathbf{U}$ and perpendicular to $\mathbf{N}$. That is the set of points $\mathbf{X} = (x, y, z)$ that satisfy

$$\mathbf{N} \cdot (\mathbf{X} - \mathbf{U}) = 0$$

For $\mathbf{U}$ and $\mathbf{N}$ specified above this equation can be written as

$$2(x - 1) + 6(y - 2) - 3(z - 3) = 0$$

or as $z = -\frac{5}{3} + \frac{2}{3}x + 2y$.



Suppose that a plane in $\mathbb{R}^3$ is given by the equation

$$\mathbf{N} \cdot (\mathbf{X} - \mathbf{U}) = 0,$$

where $\mathbf{U}$ is a point in the plane and $\mathbf{N}$ is a nonzero vector orthogonal to the plane.

The distance from a point $\mathbf{V}$ to that plane is

$$\|\mathbf{V} - \mathbf{U}\| \cos\theta,$$

where $\theta$ is the angle between $\mathbf{N}$ and $\mathbf{V} - \mathbf{U}$. In the case where $\mathbf{N}$ is a unit vector that is

$$\text{distance } = \mathbf{N} \cdot (\mathbf{V} - \mathbf{U}), \quad (\|\mathbf{N}\| = 1)$$

